

RSC Class 1

Welcome, Introduction to Microeconomics, Supply and Demand

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Agenda for Today

Welcome to Your (First) Economics Class!

Instructor: Meng Hsuan “Rex” Hsieh

Teaching Assistant/Mentor:

Today...

- Getting to Know You!
- Course Expectations
- What is Economics, and What Are Economic Costs
- Supply and Demand, Equilibrium

About Me



Course Expectations

An executive summary (1/6)

Required:

- Come to class! There are only 8 classes (class 9 is exam + wrapping up)
- Do all pre-class work as listed in Syllabus and Canvas
- Attempt all questions you see; these will appear on the exam either verbatim or with very similar wording

Course Expectations

An executive summary (2/6)

Resources:

- All slide decks I write
- Post-class slides with my annotations and in-class notes
- Me, and the fabulous
- Each other!
- An accompanying text with worked examples (on Canvas)

Course Expectations

An executive summary (3/6)

Course evaluation:

- 30% Quizzes (x3) (2 in class; 1 on Canvas)
- 20% Attendance & Participation
- 30% Final Exam (Class #8, 45~60 minutes, format TBA)
- 20% **Summarizing Highlights and Reviewing Essentials**
(SHARE)

Come to every class! This is a short course. No make-up quiz or final exam.

Course Expectations

An executive summary (4/6)

You will notice...

- No homework!
- Preparing for the 3 quizzes are the main prep you're responsible for
- I will provide practice questions to prep for quiz and final exam

Course Expectations

An executive summary (5/6)

Finally...

- **SHARE:** group “presentations”, in-class
- Groups will be selected at random every class (except class with in-class quiz)
- Every group will present at least once over the course
- I will announce the SHARE group presentations ahead of time

Course Expectations

An executive summary (6/6)

My expectations of everyone:

1. **Attend** each class.
2. **Be prepared for each class**: complete assigned readings before class; ask any clarifying questions via emails/in-person; etc.
3. **Participate** in class discussions.
4. **Communicate** any concerns or issues in real-time. **Don't wait.**
5. **Always try your best.** This'll go a long way.

Questions?

What is (Micro)economics?

+

Decision-making in Economics

What is Economics?

Microeconomics is, very broadly, the study of decision-making

- Making a decision means one alternative is selected over another $\Rightarrow$ *trade-offs* are incurred
- To understand how decisions are made, we need to analyze scarcity + choice, and opportunity cost
- Understanding these three ingredients $\Rightarrow$ we can start analyzing intriguing phenomena that we observe

This class is about learning the fundamental tools that go into economic analyses

What is Economics?

Def. A **scarce good** is one for which the **choice** of one alternative requires that another be given up

Scarcity is associated with three fundamental questions:

1. What should be produced?
2. How should goods and services be produced?
3. For whom should goods and services be produced?

What is Economics?

Def. Opportunity cost is the value of the best alternative forgone in making any choice

Example. coming to class; the opportunity costs are...

What does having a nonzero opportunity cost imply?

What is Economics?

Example. The manager of a Ford assembly plant is considering whether to produce cars or sport utility vehicles (SUVs) next month. Assume (a) they can only produce one kind of vehicle at a time, (b) quantities of labor and other materials required would be the same for either type of production.

What are the scarcity problem and opportunity costs faced by this assembly plant?

How to Make Economic Decisions?

Scarcity implies trade-offs exists: opportunity costs are (generally) nonzero

Economists assume individuals maximize the *net benefit* of each potential decision, defined as total benefit of the activity minus its opportunity costs

What do we use to characterize how these decisions are made?

How to Make Economic Decisions?

What do we use to characterize how these decisions are made?

Answer: thinking on the margin

We need to look at the additional benefit and the additional cost of another unit of the activity

Def. The marginal benefit is the amount by which an additional unit of an activity increases an individual's total benefit

Def. The marginal cost is the amount by which an additional unit of an activity increases an individual's total cost

How to Make Economic Decisions?

Decision rule: we want to make a decision where the marginal benefit equals marginal cost

Specifically, think about the marginal decision of doing an activity:

- If $MB > MC$, then $\uparrow Q$ (i.e. do more of the activity)
- If $MB < MC$, then $\downarrow Q$ (i.e. do less of the activity)

Why? Imagine buying a subscription to Spotify, an all-or-nothing decision

Another reason why we think this way: Calculus (later!)

How to Make Economic Decisions?

We will spend a lot of time with the following decision rule:

Individuals and firms make decision at which

Marginal benefit equals marginal cost ($MB = MC$)

Please remember this decision rule!

We now begin translating this fundamental decision into objects that are important in economics

Roadmap of RSC

There are two distinct parts...

Part 1 of the course is **market analysis**

Supply and demand, making decisions on the margin, equilibrium...

→ Market structure: market power, competition, implications for firms' decision-making and profits

→ Welfare analyses

Part 2 of the course is about **strategies** and decision-making with market power

Game theory: strategies, equilibrium, first-mover advantages...

→ Apply to thinking about decision-making with uncertainty in payoffs

Supply, Demand, Equilibrium

Demand

We begin with demand

Def. Demand for a good or service is the amount of good that a consumer is willing to buy at a given price

Demand

When we talk about demand, there is always a market

Def. Market refers to a specific product being bought and sold at a given quantity, at a specific location, and at a given point in time

Therefore, “my demand” always refers to a specific product or other commodities and services

My demand for a product is equivalent to saying what quantity of a product I want

What affects my demand for a product?

Demand

What affects my demand for a product?

1. Price
2. My income
3. My desires and tastes
- 4.

Demand

What affects my demand for a product?

1. Price

What is the relationship between price of a product and its quantity demanded?

Answer: law of demand

Definition:

Demand

What affects my demand for a product?

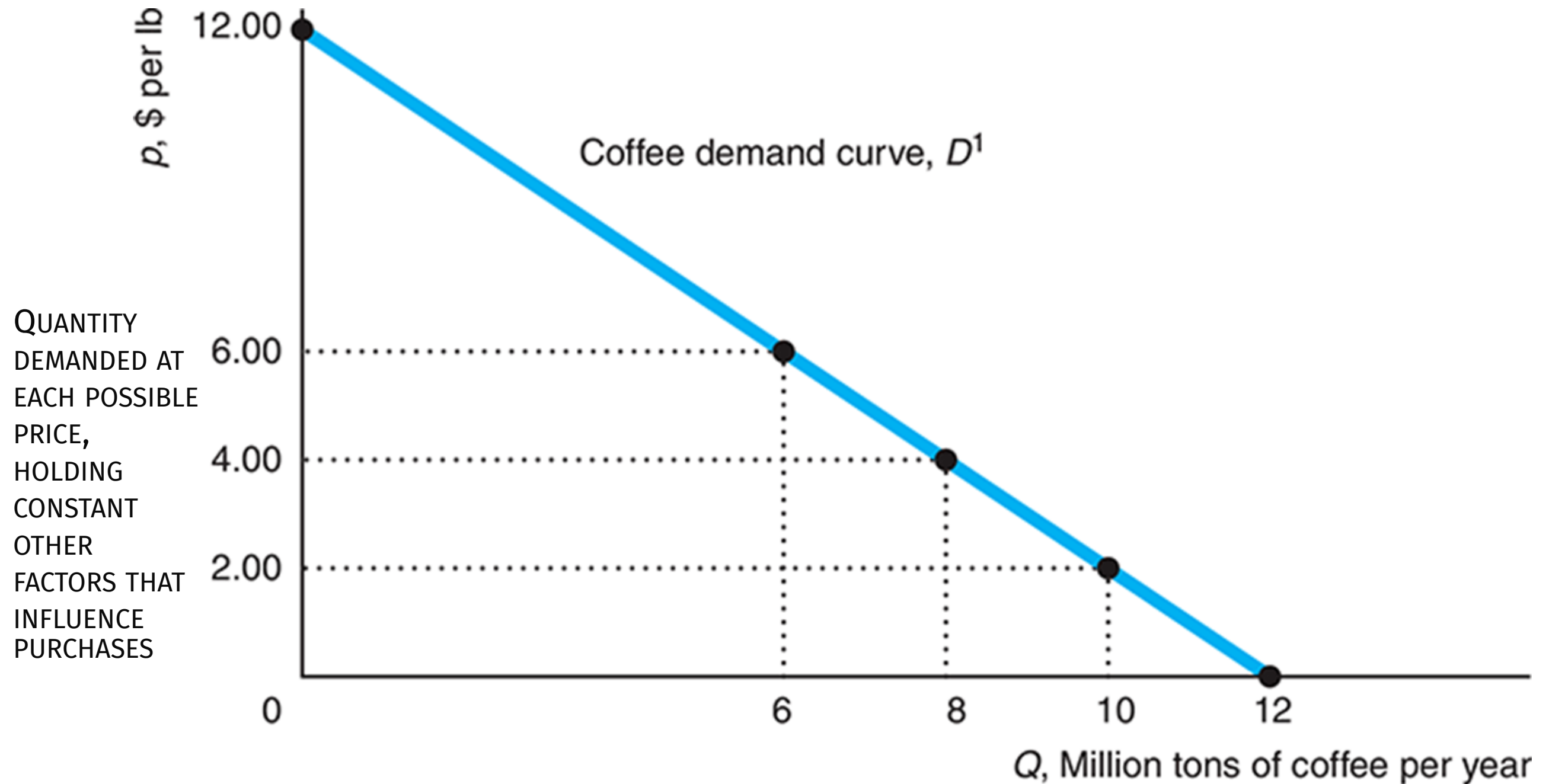
1. Price

What is the relationship between price of a product and its quantity demanded?

Answer: law of demand

How to graphically represent demand for a product?

Demand



Supply

We now define supply

Def. Supply of a product is the quantity of good that a producer is willing to (produce and) sell

It is important to emphasize “willing to sell” in the definition, and we will see this is a function of costs

What affects a producer’s supply of a product?

Supply

What affects a producer's supply of a product?

1. Price
2. Cost of production
3. Seller's outside options
- 4.

Supply

What affects a producer's supply of a product?

1. Price

What is the relationship between price of a product and its quantity demanded?

Answer: law of supply

Definition:

Supply

What affects a producer's supply a product?

1. Price

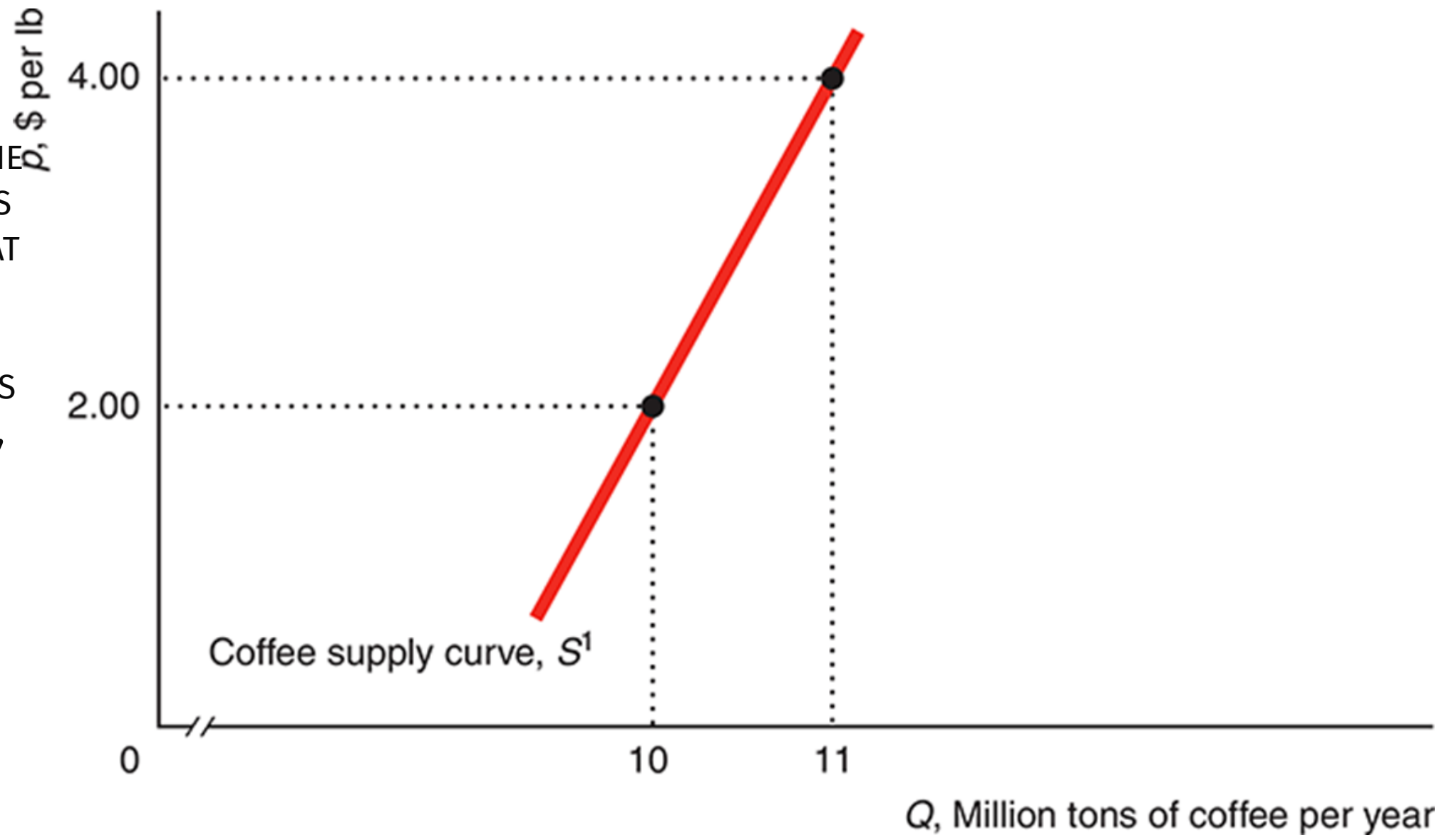
What is the relationship between price of a product and its quantity supplied?

Answer: law of supply

How to graphically represent supply of a product?

Supply

FIRMS DETERMINE SUPPLY ON BASIS OF PRICE OF THAT GOOD AND ON OTHER FACTORS, INCLUDING COSTS OF PRODUCTION, TECHNOLOGICAL CHANGE, GOVERNMENT REGULATIONS



Supply and Demand

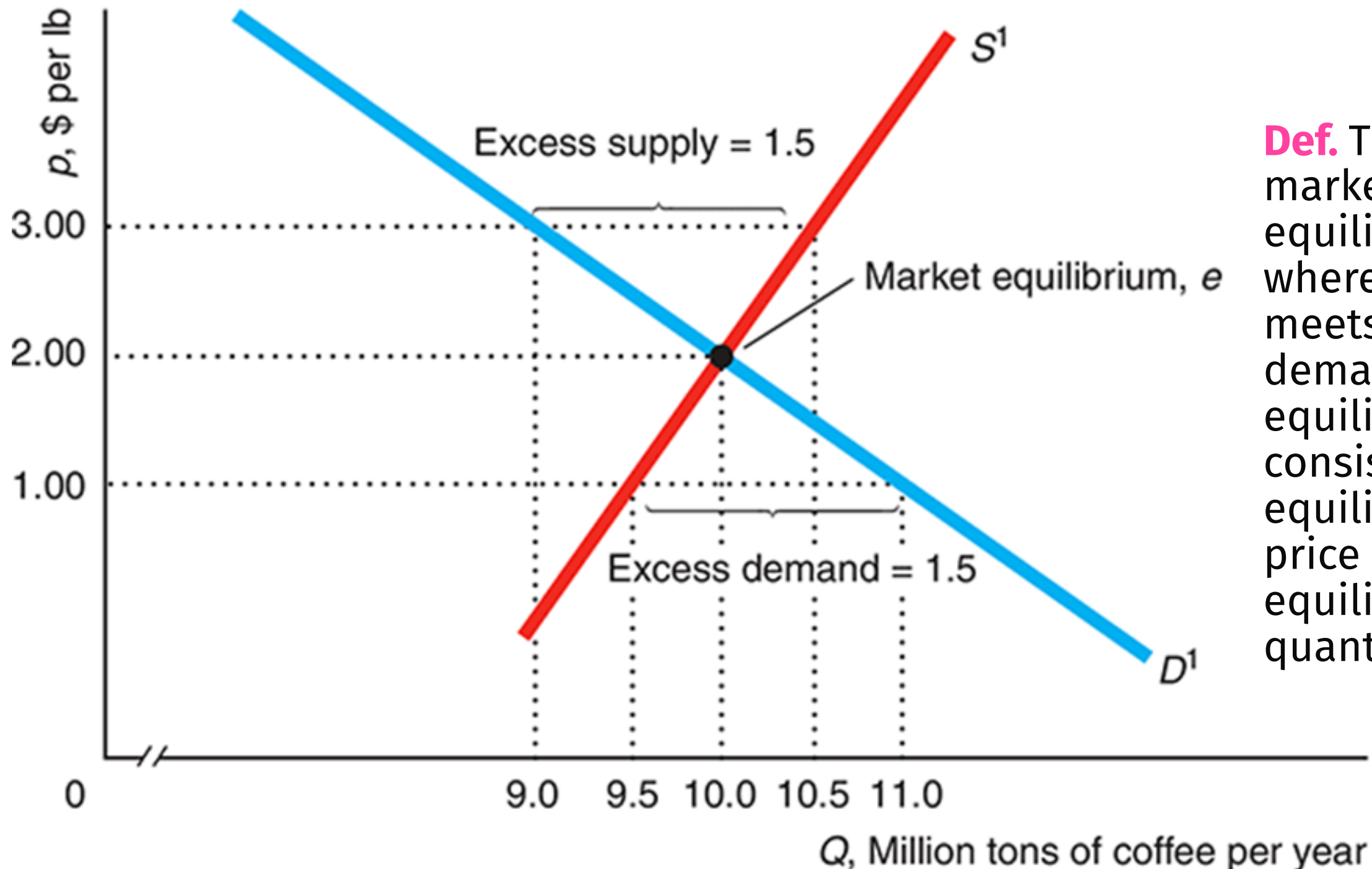
Together, supply and demand form the basic working model of modern economics

Assumptions for this model to be meaningful for analyses

1. Focuses on a single market with many buyers and sellers
2. All goods/services sold in the market are identical
3. All goods/services sold in the market are transacted at the same price.
4. Everyone has the same information about prices, quality of goods/services, etc

WE CAN NOW PUT DEMAND AND SUPPLY TOGETHER!

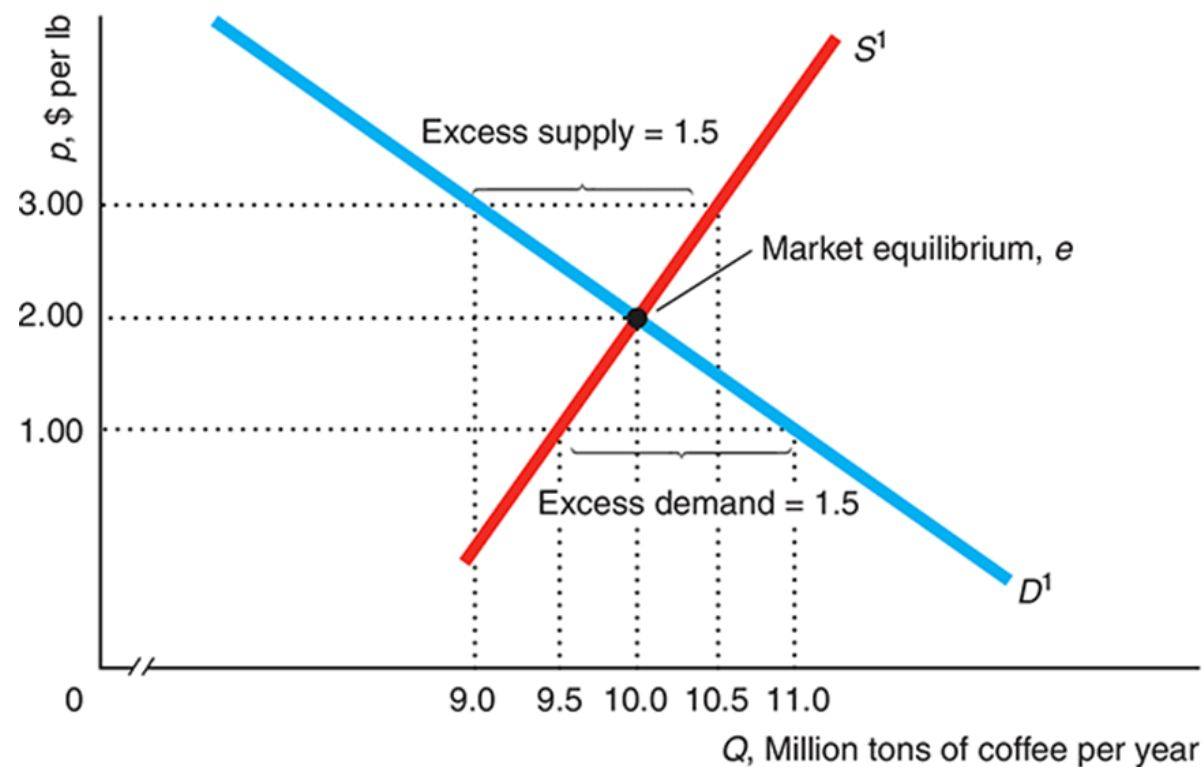
Supply and Demand



Def. The market equilibrium is where supply meets demand. Any equilibrium consists of an equilibrium price and equilibrium quantity.

Supply and Demand

We left off here



Remember markets are defined for a product, at a particular place, at a point in time

Therefore, any equilibrium describes a snapshot of individual consumers' and firms' behavior at a given time

What if there are fewer consumers?

What if firms get less raw materials for production because of extreme weather events?

Supply and Demand

Movements along the supply/demand curves

- If only the price of a product goes up (everything else held constant), then a consumer would want to consume [more / less]
- Likewise, if only the price of a product a firm can sell at goes up (everything else held constant), then a producer is willing to produce [more / less]

These statements are equivalent to movement along the supply/demand curves

These statements are also equivalent to saying ceteris paribus (ALL ELSE HELD CONSTANT, A PRICE CHANGE...)

What if there are more than price changes?

Supply and Demand

Shifts of the supply/demand curves

In life, there are many changes that are not related to price of the product changing

E.g. extreme climate events, limited supply of raw materials, shipping delays, shrinking customer base, ...

- If there are changes to any factors other than price that affect demand/supply, then the entire demand/supply curve shifts

Where to go from here?

A brief recap...

So far, we have only defined the fundamental objects of modern economic analyses. We have not said anything about how to use these fundamental objects to analyze why certain market phenomena occur

Next time, we will

1. Explain how economists use supply and demand model to analyze changes in markets
2. Work with some numerical examples
3. And then define more objects of interest within the supply and demand model, starting with **elasticity**

For next time

Supply and demand, demand elasticity, pricing

Come to office hours w/ any questions/concerns — or email me!

See you next Monday @ 9am sharp!